Empirical Predictions of the Topological Saturation Model (ΛTSM): Falsifying Dark Matter via High-Redshift Kinematics and Macroscopic Lensing

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Abstract — We present four falsifiable observational predictions of the **Topological Saturation Model (ΛTSM)**, a parameter-free covariant Dirac–Born–Infeld (DBI) field-theoretic framework demonstrated in our foundational studies [1, 2]. The model derives early-universe Cosmic Microwave Background (CMB) acoustic spectra, galactic rotation curves, and strong gravitational lensing solely from visible baryonic mass and the causal Hubble horizon acceleration floor $a_0 = \frac{cH_0}{2\pi} \approx 1.128 \times 10^{-10}$ m s^{-2}, rendering non-baryonic dark matter mathematically redundant. **Prediction I (JWST High-Redshift Kinematics):** Dynamically settled, thin rotating disk galaxies ($\eta = b/a \to 0$) at $z > 3$ observed by the James Webb Space Telescope (JWST) will immediately exhibit flat outer rotation curves obeying $V_{\text{calculated}} = \sqrt{V_{\text{baryonic}}^2\, \nu}$ with a planar Kolmogorov–Arnold–Moser (KAM) golden-ratio threshold $a_\varphi = a_0\varphi \approx 1.826 \times 10^{-10}$ m s^{-2} and a baseline stellar mass-to-light ratio $\Upsilon_* \approx 0.46$, whereas unrelaxed or thick turbulent proto-disks ($\eta \to 1$) will exhibit Newtonian-like decline via the geometric gate $a_{0,\text{eff}} = a_0\sqrt{1-\eta^2}$. **Prediction II (Euclid Galaxy Clusters):** Morphologically relaxed, single-node galaxy clusters mapped by the Euclid mission will yield physical Einstein ring radii dictated by the Macroscopic Saturation Boundary $R_{BB} = \sqrt{GM_{\text{baryon}}/a_0}$ using strictly the observable $\sim 13\%$ baryonic mass fraction (X-ray gas and stellar mass) with 0% dark matter, matching the $R^2_{1:1} = 0.9710$ ($p \approx 1.4 \times 10^{-11}$) 1:1 parity established across relaxed CLASH clusters. **Prediction III (Euclid and Roman Isolated Strong Lenses):** Tens of thousands of newly discovered early-type strong lenses will obey the parameter-free relativistic DBI waveguide deflection angle $\hat{\alpha} = \frac{4\pi\sqrt{GM_{\text{baryon}}a_0}}{c^2}$ and physical Einstein radius $R_{E,\text{calc}} = D_L \frac{D_{LS}}{D_S}\hat{\alpha}$, reproducing the 1.016 mean observed-to-predicted ratio (1.6% mean parity) and 98.6% ensemble sum parity verified across $N = 110$ Grade-A SLACS and S4TM lenses. **Prediction IV (Roman Sub-Arc Perturbers):** Ultra-high-resolution lensing observations by the Nancy Grace Roman Space Telescope and ALMA/VLBI interferometry will confirm that localized breaks, quadruplet vertices, and millilensing perturbations along Einstein rings map deterministically to compact baryonic structures—giant molecular clouds (GMCs), star-forming clumps, and wandering black holes—rather than diffuse Navarro–Frenk–White (NFW) dark matter subhalos.

Highlights

- Formulates four falsifiable near-term tests of the zero-parameter ΛTSM framework across JWST, Euclid, and Roman.
- Predicts immediate rotational flattening in thin $z > 3$ disks ($\eta \to 0$) vs. Newtonian decline in thick spheroids ($\eta \to 1$).
- Establishes parameter-free cluster lensing parity ($R_{BB} = \sqrt{GM_{\text{baryon}}/a_0}$) using only the $\sim 13\%$ baryonic mass budget.
- Predicts 1 : 1 relativistic waveguide Einstein ring scaling across isolated Euclid and Roman galaxy-scale lenses.
- Attributes sub-arc lensing anomalies to compact baryonic GMCs and star-forming clumps rather than dark subhalos.

Keywords: Dark matter falsification; Topological Saturation Model (ΛTSM); Dirac–Born–Infeld field theory; JWST high-redshift kinematics; Strong gravitational lensing (Euclid, Roman, SLACS, CLASH).

1 Introduction

The standard ΛCDM cosmological paradigm interprets flat galactic rotation curves and strong gravitational lensing as indirect evidence for non-baryonic Cold Dark Matter (CDM) halos [3–6]. Despite over two decades of intensive direct, collider, and multi-messenger indirect searches across gamma-ray, neutrino, cosmic-ray, and X-ray channels, no dark matter particle has achieved discovery-level significance [6, 7]. Conversely, classical empirical modifications to inertia [8–10] struggle to simultaneously account for early-universe Cosmic Microwave Background (CMB) acoustic oscillations, ultra-diffuse Newtonian galaxies, and colliding galaxy clusters.

In our foundational companion studies [1, 2], we demonstrated that these phenomena are deterministic emergent properties of a covariant four-dimensional Dirac–Born–Infeld (DBI) phase transition in the spatial manifold—the **Topological Saturation Model (ΛTSM)** [11, 12]. In the early-universe ultra-relativistic regime ($2|X| \to M^4$), the DBI action yields three exact parameter-free constants from first principles [2]: (1) the 4D orthonormal tetrad background

density ratio $\omega_{\text{phantom}}/\omega_b = \frac{16}{3}\left(1+\frac{1}{16\pi^2}\right) \approx 5.367107$, fixing $\omega_{\text{phantom}} = 0.120062$ directly from the baryon density $\omega_b = 0.02237$; (2) the 3D Maclaurin curvature elasticity $\mathcal{K}_0 = \frac{2}{3}$; and (3) the covariant Einstein–DBI horizon impedance $\Gamma_0 = \frac{1}{8\pi}$, reproducing the Planck 2018 CMB power spectra ($R^2_{TT} = 0.9991, R^2_{TE} = 0.9954, R^2_{EE} = 0.9957$) and linear matter clustering ($\sigma_8 = 0.8248 \pm 0.0055$) with 0% CDM [2].

In the late-universe macroscopic regime, coupling the causal propagation speed (c) to the background Hubble expansion rate ($H_0 \approx 73.0$ km s^{-1} Mpc^{-1} in the unperturbed Hubble flow $D > 20$ Mpc [13]) via causal horizon closure [14–16] imposes a rigid minimal acceleration floor [1]:

$$a_0 = \frac{cH_0}{2\pi} = \frac{c^2}{2\pi R_H} \approx 1.128 \times 10^{-10} \text{ m s}^{-2} \tag{1}$$

Crucially, the activation of this topological boundary is strictly conditioned on two physical gates [1]:

1. **The Morphological Flattening Gate ($\eta = b/a$):** While external relativistic null geodesics ($ds^2 = 0$) project onto the 2D sky-plane aperture independently of internal 3D thickness, internal sub-relativistic orbital dynamics ($v \ll c$) are governed by the effective threshold:

$$a_{0,\text{eff}} = a_0\sqrt{1-\eta^2} \tag{2}$$

   where b is vertical thickness and a is radial diameter. In flat spiral disks ($\eta \to 0$), $\sqrt{1-\eta^2} \to 1$ activates full kinematic saturation, whereas in uncompressed 3D spheroids ($\eta \to 1$) such as ultra-diffuse dwarf galaxies (e.g., NGC 1052-DF2 [17]), $a_{0,\text{eff}} \to 0$ restores purely Newtonian decay.

2. **The Virial Relaxation Gate (Θ_{relax}):** Cohesive boundary formation requires a thermodynamically relaxed, single-node gravitational potential ($N_{\text{nodes}} = 1$). In unrelaxed multi-node interactions ($N_{\text{nodes}} > 1$), such as the Milky Way–Andromeda binary infall separated by $d \approx 0.77$ Mpc (where the mutual peculiar infall velocity $v_{\text{pec}} \approx 110$ km s^{-1} doubles the local Hubble expansion flow $v_H = H_0 \times D \approx 56.2$ km s^{-1}) [18, 19] or violent cluster collisions [20, 21], the intermediate manifold undergoes contraction rather than expansion, shutting down the saturation envelope ($\Theta_{\text{relax}} \to 0$). This multi-node boundary shutdown is directly corroborated by *Gaia* Data Release 3 (DR3) kinematics demonstrating that the Milky Way's outer rotation curve transitions into a Keplerian-like decline toward the Magellanic Clouds and the Local Group boundary [22].

To empirically distinguish ΛTSM from flexible dark matter halo models across ongoing and upcoming flagship surveys, we formulate four rigorous, parameter-free predictive bounds.

2 Prediction I: Baryon-Only Rotational Flattening in High-Redshift Galaxies (JWST)

2.1 Standard ΛCDM Paradigm

Standard hierarchical structure formation predicts that galaxies at high redshift ($z > 3$) reside within still-assembling dark matter halos characterized by high merger rates and low concentration parameters. Consequently, ΛCDM models anticipate either declining outer rotation curves or strong redshift-dependent evolution in the Baryonic Tully–Fisher Relation (BTFR) [23, 24], requiring individual halo mass and concentration fitting per epoch.

2.2 The ΛTSM Prediction

Under ΛTSM, rotational flattening is an immediate geometric consequence of planar dimensional reduction ($\eta \to 0$) paired with golden-ratio Kolmogorov–Arnold–Moser (KAM) orbital resonance stability ($\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$) along logarithmic spiral arms [1, 25]:

$$a_\varphi = a_0\varphi \approx 1.826 \times 10^{-10} \text{ m s}^{-2} \tag{3}$$

For any galaxy with observable neutral gas (V_{gas}), stellar disk (V_{disk}), and stellar bulge (V_{bulge}) components at galactocentric radius r (in kpc), the total baryonic circular velocity and classical Newtonian acceleration (in SI units of m s^{-2}) are explicitly defined as [1, 26]:

$$V_{\text{baryonic}}(r) = \sqrt{V^2_{\text{gas}} + \Upsilon_* V^2_{\text{disk}} + \Upsilon_* V^2_{\text{bulge}}} \tag{4}$$

$$a_{\text{baryonic}}(r) = \frac{(1000\,V_{\text{baryonic}})^2}{3.086 \times 10^{19}\,r} \tag{5}$$

As $a_{\text{baryonic}}(r)$ decays toward a_φ, the first integral of the static DBI Euler–Lagrange field equation ($\nabla \cdot [\mathcal{L}_X \nabla\phi] = 4\pi G\rho_{\text{baryon}}$ with dielectric susceptibility $\mathcal{L}_X = \nu^{-1}$) induces the covariant topological deformation factor [1]:

$$\nu(r) = \frac{1}{1 - \exp\left(-\sqrt{\frac{a_{\text{baryonic}}(r)}{a_\varphi}}\right)} \tag{6}$$

The resulting rotational velocity profile is deterministically governed by:

$$V_{\text{calculated}}(r) = \sqrt{V^2_{\text{baryonic}}(r)\,\nu(r)} \tag{7}$$

In the deep low-acceleration periphery ($a_{\text{baryonic}} \ll a_\varphi$), first-order expansion yields $\nu \approx \sqrt{a_\varphi/a_{\text{baryonic}}}$, analytically canceling radial dependence (r) and locking the outer disk into the parameter-free asymptotic BTFR plateau [1, 23, 24]:

$$V_\infty = (GM_{\text{baryon}}a_\varphi)^{1/4} \tag{8}$$

Crucially, because $V_\infty \propto a_\varphi^{1/4} \propto H_0^{1/4}$, the $\sim 8\%$ difference between late-universe ($H_0 \approx 73.0$ km s^{-1} Mpc^{-1}) and early-universe CMB ($H_0 = 67.36$ km s^{-1} Mpc^{-1}) expansion baselines shifts predicted asymptotic rotation velocities by less than 1.9% [1,2]. Furthermore, because the primordial DBI curvature elasticity ($\mathcal{K}_0 = \frac{2}{3}$) and horizon impedance ($\Gamma_0 = \frac{1}{8\pi}$) establish gravitational potential wells prior to recombination [2], dynamically settled disks at $z > 3$ lock immediately into flat rotation curves without awaiting dark matter halo assembly. Across all 175 local SPARC galaxies ($N = 3,391$ radial points) [26], this exact law achieves $R^2 = 0.9150$ ($p < 10^{-300}$) under a universally locked 3.6 μm mass-to-light ratio $\Upsilon_* = 0.46$ [27, 28] and $R^2 = 0.9698$ within standard stellar population synthesis bounds ($\Upsilon_* \in [0.1, 1.0]$) [1, 29]. A comprehensive comparison between standard ΛCDM halo expectations and the parameter-free kinematic predictions of ΛTSM for high-redshift galaxies is presented in Table 1.

Table 1: Prediction I — Kinematic Falsification in High-Redshift Galaxies ($z > 3$, JWST).

Kinematic Observable	Standard Model (ΛCDM) Expectation	Topological Saturation Model (ΛTSM) Prediction
Outer Rotation Curve ($z > 3$)	Evolving or Keplerian-declining profiles awaiting hierarchical dark halo assembly.	Immediate asymptotic flattening ($V_\infty = (GM_{\rm baryon} a_\varphi)^{1/4}$) in all settled thin disks ($\eta \to 0$).
Morphological Dependence	Halo dominance assumed across both thin disks and spheroids.	Strict gating via $a_{0,\rm eff} = a_0\sqrt{1-\eta^2}$: flat in thin disks ($\eta \to 0$), Newtonian decline in thick proto-disks ($\eta \to 1$).
Rotational Velocity Law	Requires 2–3 free NFW halo parameters per galaxy per epoch.	Deterministic: $V_{\rm calculated} = \sqrt{V^2_{\rm baryonic}\,\nu}$ with zero free dark parameters.
Stellar Mass-to-Light Ratio	Treated as a free parameter degenerate with halo concentration.	Locked to population synthesis baseline ($\Upsilon_* \approx 0.46$, or bounded within $\Upsilon_* \in [0.1, 1.0]$).
Empirical Benchmark ($z = 2.64$)	Requires massive dark halo + ad-hoc AGN feedback tuning in MRG-M0150.	Reproduces MRG-M0150 rotation velocity at $r_{\rm out} = 2.3R_e = 3.3$ kpc ($V_{\rm calc} = 453.3$ km s^{-1} vs. $v_{\rm rot} \approx 450$ km s^{-1}, 0.7% parity).

2.3 Empirical Falsification Test

As summarized in Table 1, JWST/NIRSpec IFU and NIRCam slitless grism surveys of high-redshift galaxies ($z > 3$) will confirm three falsifiable kinematic signatures:

- Dynamically settled, cold rotating disks ($v/\sigma > 2, \eta \to 0$)—already identified at $z > 3$ [30, 31], out to $z \approx 7.3$–14.2 [32], and in settled disk components at $z = 10.231$ (GHZ8 [33])—will obey Eqs. 6–8 with zero dark matter parameters.
- In massive quiescent fast-rotators at Cosmic Noon such as MRG-M0150 ($z = 2.64, \log(M_*/M_\odot) = 11.30 \pm 0.04, < 1\%$ molecular gas fraction, $R_e \approx 1.6$ kpc) [34], evaluating Eqs. 6–7 at the outer JWST/NIRSpec IFU boundary $r_{\rm out} = 2.3R_e = 3.3$ kpc (where the enclosed Sérsic baryonic mass $M_{\rm enc} \approx 0.76 M_* \approx 1.52 \times 10^{11}\,M_\odot$ yields $V_{\rm baryonic} = 444.5$ km s^{-1} and $a_{\rm baryonic} = 1.94 \times 10^{-9}$ m s$^{-2} \gg a_\varphi$, so $\nu = 1.040$) predicts $V_{\rm calculated} = 453.3$ km s^{-1}, matching the observed stellar rotation velocity ($v_{\rm rot} \approx 450$ km s^{-1}) to within 0.7%, while non-virialized bi-conical AGN outflows (1000–1400 km s^{-1}) decouple perpendicular to the disk in accordance with $\Theta_{\rm relax}$ [34, 35].
- At intermediate redshifts ($0.1 < z < 1.43$), inner mass density slopes will remain governed exclusively by baryonic stellar mass (M_*) independent of supernova feedback burstiness [36–39].

3 Prediction II: Macroscopic Lensing Boundaries in Uncharted Galaxy Clusters (Euclid)

3.1 Standard ΛCDM Paradigm

Standard cosmology anticipates that newly discovered galaxy clusters mapped by the Euclid mission [40, 41] will require multi-component dark matter halo reconstructions (often injecting up to 21 free halo and subhalo parameters per cluster [34, 42]) because visible baryons account for only $\sim 13\%$ of the dynamical mass inferred under unconstrained General Relativity.

3.2 The ΛTSM Prediction

In any morphologically relaxed, single-node system ($\Theta_{\rm relax} = 1$), the external radial baryonic acceleration gradient ($a = GM_{\rm baryon}/r^2$) intersects the cosmological horizon floor a_0 at the **Macroscopic Saturation Boundary** (R_{BB}) [1]:

$$R_{BB} = \sqrt{\frac{GM_{\rm baryon}}{a_0}} \tag{9}$$

At $r = R_{BB}$, the spatial metric undergoes an optical phase transition into a 2D thin-lens waveguide sheath, routing background null geodesics along the perimeter. Across the Cluster Lensing And Supernova survey with Hubble (CLASH) [43, 44], evaluating Eq. 9 using strictly the observed $\sim 13\%$ baryonic mass fraction (intracluster X-ray gas plus bright cluster galaxy stellar mass) with 0% dark matter reproduces the observed physical Einstein ring radii of all morphologically relaxed clusters with a strict 1:1 identity parity of $R^2_{1:1} = 0.9710$ ($p \approx 1.4 \times 10^{-11}$) [1]. The contrasting theoretical assumptions, mass budget requirements, and quantitative lensing benchmarks between ΛCDM and ΛTSM at the galaxy-cluster scale are summarized in Table 2.

3.3 Empirical Falsification Test

In accordance with the criteria outlined in Table 2, Euclid strong and weak lensing surveys of uncharted galaxy clusters [40, 41], alongside stacked DESI DR2 + ACT DR6 kinematic Sunyaev–Zel'dovich (kSZ) and CMB-lensing profiles [45], will confirm two falsifiable signatures:

1. **Relaxed Cluster 1:1 Parity:** For all virially relaxed, single-node clusters ($0.85 < \eta_{\rm vir} < 1.15$, centroid offset $\Delta_r < 0.04$ [21]), the observed physical Einstein ring radius will match $R_{BB} = \sqrt{GM_{\rm baryon}/a_0}$ ($R^2_{1:1} > 0.95$) using solely the observable $\sim 13\%$ baryonic mass. For a benchmark cluster baryonic payload of $M_{\rm baryon} = 0.98 \times 10^{13}\,M_\odot$, the metric deterministically locks the lensing boundary at $R_{BB} = 110.0$ kpc (scaling exactly as $M^{1/2}_{\rm baryon}$ from the isolated galaxy benchmark of 11.00 kpc at $M_{\rm baryon} = 0.98 \times 10^{11}\,M_\odot$). Beyond the strong-lensing core, continuous weak-lensing shear profiles out to 2–5 Mpc (such as Abell 2390 [41]) will trace the extended ionized circumgalactic and intragroup baryonic gas ($\tilde{f}_{\rm gas}(\Omega_m/\Omega_b) \to 1.0$, $\Omega_b = 0.0488 \pm 0.0064$ [45, 46]) via $\nu(r)$.
2. **Unrelaxed Merger Rupture:** Violent multi-node mergers ($\eta_{\rm vir} > 1.2, \Delta_r > 0.2$, such as MACS J0717, MACS J2129, and Bullet Cluster analogues [20, 21, 47]) will

Table 2: Prediction II — Macroscopic Lensing Boundaries in Uncharted Galaxy Clusters (Euclid).

Cluster Lensing Parameter	Standard Model (ΛCDM) Expectation	Topological Saturation Model (ΛTSM) Prediction
Photon Deflection Mechanism	Continuous 3D deflection through multi-parameter NFW dark matter halos.	Deterministic geodesic routing along the 2D Macroscopic Saturation Boundary (R_{BB}).
Cluster Einstein Radius Law	Requires up to 21 free halo/subhalo parameters per cluster reconstruction.	Parameter-free: $R_E = R_{BB} = \sqrt{GM_{\rm baryon}/a_0}$ for relaxed clusters ($f_b \approx 13\%$).
Mass Input Required	$\sim 87\%$ invisible non-baryonic dark matter + $\sim 13\%$ baryons.	Strictly the observable $\sim 13\%$ baryonic mass (X-ray gas + stellar mass); 0% dark matter.
Relaxed vs. Merging Clusters	Predicts large lensing cross-sections in both relaxed and merging halos.	Strict virial gating ($\Theta_{\rm relax}$): $R^2_{1:1} > 0.95$ in relaxed clusters (matching CLASH $R^2_{1:1} = 0.9710$), vs. envelope rupture in violent multi-node mergers.
Numerical Scale Benchmark	Unpredictable without post-hoc dark halo fitting.	For a benchmark cluster baryonic mass $M_{\rm baryon} = 0.98 \times 10^{13}\,M_\odot$, locks $R_{BB} = 110.0$ kpc (11.00 kpc at galaxy scale $0.98 \times 10^{11}\,M_\odot$) with 0 free parameters.

exhibit ruptured global envelopes and an order-of-magnitude collapse in coherent single-envelope Einstein radii, while individual virialized member galaxies retain their local saturation sheaths outside the shocked X-ray gas bridge.

4 Prediction III: Strong Lensing in Isolated Galaxies (Euclid and Roman)

4.1 Standard ΛCDM Paradigm

Standard lens modeling assumes that early-type elliptical galaxies are embedded within extended NFW or power-law dark matter halos [48]. However, recent spatially resolved JWST and Keck kinematic audits of SLACS deflectors demonstrate that standard power-law halo models break down at the population level when $R_{\rm eff} \approx R_E$, requiring an ad-hoc internal mass-sheet transformation parameter ($\lambda_{\rm int} = 1.12^{+0.05}_{-0.06} \neq 1$) with intrinsic scatter [42].

4.2 The ΛTSM Prediction

For cosmological lens-source configurations satisfying the waveguide capture criterion ($D_{LS}/D_S \gtrsim 0.5$), integrating the logarithmic DBI saturation potential $\Phi_{\rm sat}(r) = V_\infty^2 \ln(r/R_{BB})$ (where $V_\infty^4 = GM_{\rm baryon}a_0$) along the line of sight yields the parameter-free relativistic deflection angle [1]:

$$\hat{\alpha} = \frac{4\pi V_\infty^2}{c^2} = \frac{4\pi\sqrt{GM_{\rm baryon}a_0}}{c^2} \tag{10}$$

Projecting Eq. 10 onto the physical lens plane determines the predicted Einstein ring radius:

$$R_{E,\rm calc} = D_L \frac{D_{LS}}{D_S}\hat{\alpha} = D_L \frac{D_{LS}}{D_S}\frac{4\pi\sqrt{GM_{\rm baryon}a_0}}{c^2} \tag{11}$$

Because the optical phase transition occurs sharply at R_{BB}, the apparent departure from a continuous 3D power-law profile at $R_{\rm eff} \approx R_E$ emerges naturally with zero free parameters. Across the combined catalog of $N = 110$ confirmed Grade-A early-type lenses from SLACS IX ($N = 70$ complete multi-band HST lenses [49–52]) and SLACS XIII S4TM ($N = 40$ lower-mass lenses [53]), Eqs. 10–11 achieve $R^2 = 0.8261$ ($p \approx 1.6 \times 10^{-27}$, strict 1:1 parity $R^2_{1:1} = 0.8156$) across SLACS IX and $R^2 = 0.7955$ ($p \approx 5.1 \times 10^{-39}$, $R^2_{1:1} = 0.7773$) across the combined $N = 110$ dataset, with a mean observed-to-predicted ratio of $\langle R_{E,\rm obs}/R_{E,\rm calc}\rangle = 1.016 \pm 0.198$ (median 1.028, i.e., 1.6% mean parity) and an ensemble sum of 370.3 kpc observed versus 375.7 kpc predicted (98.6% ensemble parity) [1]. Table 3 summarizes the falsification framework contrasting retroactive ΛCDM halo-plus-mass-sheet modeling against the zero-parameter relativistic waveguide predictions of ΛTSM for isolated galaxy-scale lenses.

4.3 Empirical Falsification Test

Upcoming wide-field surveys by the Euclid mission, the Nancy Grace Roman Space Telescope, and the Korean Space Telescope (KST) [40, 47] are projected to discover $> 10^5$ new galaxy-scale strong gravitational lenses. As detailed in Table 3, we predict that across all newly discovered isolated, relaxed early-type lenses, the observed physical Einstein ring radii ($R_{E,\rm obs} = D_L\theta_E$) will follow Eqs. 10–11 directly from published stellar population synthesis masses ($M_{\rm baryon}$) and spectroscopic redshifts (z_L, z_S) with zero dark matter parameters, maintaining ~ 98–100% ensemble sum parity and residual scatter bounded strictly by the natural ~ 0.10 dex ($\sim 25\%$) stellar population synthesis uncertainty.

5 Prediction IV: Topological Fragmentation of Lensing Rings and Baryonic Substructure (Roman and ALMA/VLBI)

5.1 Standard ΛCDM Paradigm

When high-resolution imaging reveals fragmented Einstein rings, quadruplet configurations (e.g., Einstein Crosses), or localized millilensing flux-ratio and astrometric anomalies along lensed arcs, standard cosmology attributes these perturbations to unseen populations of 10^6–$10^9\,M_\odot$ collisionless dark matter subhalos residing within the primary lens halo [42, 48].

5.2 The ΛTSM Prediction

Under ΛTSM, the optical waveguide perimeter at R_{BB} is directly sculpted by the underlying baryonic stress-energy tensor $T^{(b)}_{\mu\nu}$. In non-axisymmetric or barred potentials, null geodesics traversing R_{BB} extremize optical action along the stable topological vertices of the boundary envelope, natively generating quadruplet Einstein Crosses [1]. Furthermore, localized astrometric perturbations and optical breaks along lensed arcs are

Table 3: Prediction III — Isolated Galaxy-Scale Strong Lensing in Next-Generation Surveys (Euclid / Roman).

Galaxy Lensing Observable	Standard Model (ΛCDM) Expectation	Topological Saturation Model (ΛTSM) Prediction
Deflection Angle Formulation	Retroactive SIE/NFW halo fitting + ad-hoc mass-sheet parameter ($\lambda_{\rm int} \neq 1$).	Exact covariant DBI waveguide angle: $\hat{\alpha} = \frac{4\pi\sqrt{GM_{\rm baryon}a_0}}{c^2}$.
Physical Einstein Ring Radius	Requires dark halo mass, concentration, and anisotropy priors per lens.	Parameter-free: $R_{E,\rm calc} = D_L \frac{D_{LS}}{D_S} \frac{4\pi\sqrt{GM_{\rm baryon}a_0}}{c^2}$.
Free Dark Matter Parameters	Multiple per lens galaxy.	Strictly zero (0% dark matter).
Population Parity Benchmark	Breakdown of power-law profiles at $R_{\rm eff} \approx R_E$ requiring manual corrections.	Matches the $N = 110$ SLACS + S4TM benchmark: $\langle R_{E,\rm obs}/R_{E,\rm calc}\rangle = 1.016$ (1.6% mean parity) and 98.6% ensemble sum parity within ~ 0.10 dex SPS bounds.

produced deterministically by compact, high-surface-density baryonic structures intersecting the R_{BB} sheath.

5.3 Empirical Falsification Test

Ultra-high-resolution optical/near-infrared surveys by the Roman Space Telescope and sub-millimeter/radio interferometry (ALMA, VLBI) will falsify the dark subhalo hypothesis via three direct observables:

1. **Compact Over-Concentration vs. Diffuse NFW Profiles:** Sub-arc gravitational perturbers ($\sim 10^6 M_\odot$ enclosed within $\sim$ 10–140 pc) will consistently exhibit ultra-steep, compact surface densities ($\Sigma \sim 10^3$–$10^4 M_\odot$ pc^{-2}) that severely violate diffuse NFW dark matter subhalo predictions, matching instead dust-obscured giant molecular clouds (GMCs), compact star-forming clumps, globular clusters, and wandering black holes—a tension already emerging in recent high-resolution interferometric lensing audits [42, 54, 55].
2. **Direct Baryonic Clump Coincidence:** Multi-wavelength JWST and Roman spectroscopy will confirm that morphological breaks and millilensing anomalies along lensed arcs correlate spatially with patchy, off-center baryonic star-forming clumps ($\lesssim$ 215 pc) and dense molecular gas complexes observed across $0.5 < z < 7$ [35, 55, 56].
3. **Chromatic Extinction and Microlensing Separation:** Multi-band optical-to-infrared flux-ratio audits will demonstrate that residual flux anomalies decouple into chromatic dust extinction and stellar microlensing rather than achromatic dark subhalos [47].

6 Conclusion

By replacing unconstrained per-halo curve fitting with a covariant, zero-parameter DBI phase transition governed by the causal Hubble horizon floor ($a_0 = cH_0/2\pi$), the **Topological Saturation Model (ΛTSM)** provides a unified, strictly falsifiable roadmap across cosmic scales [1, 2]. Whether testing the immediate rotational flattening of $z > 3$ thin disks with JWST, the 1:1 baryonic saturation boundary (R_{BB}) of relaxed galaxy clusters with Euclid, the relativistic waveguide scaling of isolated strong lenses with Roman, or the baryonic GMC origin of sub-arc lensing perturbations with ALMA/VLBI, ΛTSM demands that gravitational anomalies map deterministically to their observable baryonic payload with 0% dark matter.

CRediT Authorship Contribution Statement

Tomer Haimovich: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and Code Availability

The complete observational audit datasets, Python verification scripts, modified `CLASS v3.4.0` C source code, and covariant field-theoretic Supplementary Information underlying the ΛTSM framework are permanently archived in Zenodo at https://doi.org/10.5281/zenodo.24098753 (Foundational Study Part I), https://doi.org/10.5281/zenodo.24098774 (Companion Cosmological Study Part II), https://doi.org/10.5281/zenodo.24154676 (Empirical Predictions Study), and on GitHub at https://github.com/Tomer-Haimovich/The-Phantom-Metric.

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